KU LEUVEN

DEPARTMENT OF COMPUTER SCIENCE

Compress Yourself

Amin Faghih, Kobe Bergmans, Michele Rinelli and Raf Vandebril
KU Leuven, NUMA
amin.faghih@kuleuven.be kobe.bergmans@kuleuven.be, michele.rinelli@kuleuven.be, raf.vandebril@kuleuven.be

What is an Image?

- An **Image** is a grid of pixels, where each pixel is associated to a color
- In the **RGB** format, every pixel has a value (between 0 and 255) for the Red, Green, and Blue channel, e.g.,
  - Red: (R=255,G=0,B=0), Green: (R=0,G=255, B=0), Blue: (R=0,G=0,B=255)
  - Orange: (R=255,G=165,B=0)
  - Violet: (R=238,G=130,B=238)

  A single pixel requires thus 24 bits = 3 bytes of memory.
- The **Resolution** stands for the number of pixels.
  Popular examples are
  - 640×480 (VGA), e.g., early TVs and webcams
  - 1280×720 (HD), e.g., mid 2000s TVs
  - 1920x1080 (Full HD), e.g., most nowadays TVs, monitors, smartphones
  - 2560x1440 (2K QHD), e.g., Gaming monitors, high-end phones
  - 3840x2160 (4K UHD), e.g., High-end TVs, monitors
- Without compression, the storage required for an image is #Pixels · 3 bytes
  - E.g., a picture in Full HD requires 1920·1080·3 bytes=6,220,800 bytes ≈ 6 MB.

Compression Techniques

Goal: Reduce the file size of the image, while preserving the quality

- Common techniques
  - JPEG (quantization, discrete cosine transform)
  - PNG (deflation, lossless)
  - Vector Graphics, e.g., PDF, SVG, PS, EPS (algebraic curves, scalable)
- More advanced techniques
  - Tensor decompositions
  - Wavelet transforms
  - Machine learning
- Here, we consider one of the first techniques, based on the SVD.

Technical – Singular Value Decomposition (SVD)

- Every $m \times n$ matrix A has a **Singular Value Decomposition** (SVD):

$$A = U\Sigma V^T,$$

  - $U = [\boldsymbol{u}_1, \dots, \boldsymbol{u}_m]$ is an orthogonal $m \times m$ matrix (left singular vectors)
  - $V = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_n]$ is an orthogonal $n \times n$ matrix (right singular vectors)
  - Σ is the $m \times n$ diagonal matrix of singular values, $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0$
- Reveals many important mathematical properties of a matrix
  - The rank of A is the number of nonzero singular values
  - The left/right singular vectors reveal the image/kernel of A
  - Reveals the spectral norm and condition number, and the Frobenius norm
- The **Truncated SVD** provides us the matrix $A_k = U_k \Sigma_k V_k^T$, where we have thrown away all the singular values beyond σ_k

$$U_k = [\boldsymbol{u}_1, \dots, \boldsymbol{u}_k], \quad V_k = [\boldsymbol{v}_1, \dots, \boldsymbol{v}_k], \quad \Sigma_k = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_k \end{bmatrix}$$

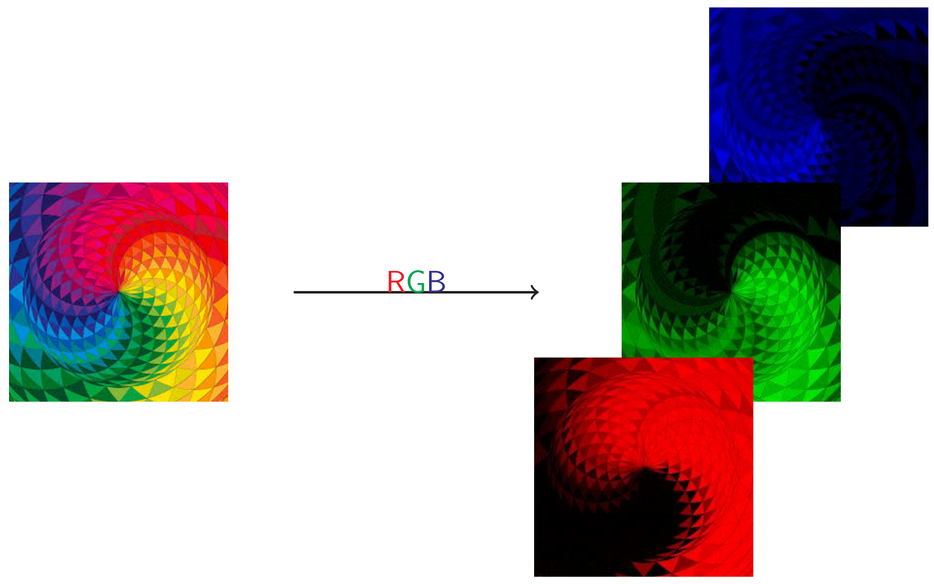
  - The matrix A_k is the compressed version of the matrix A.
    It requires much less storage.
  - It is the best rank-k approximation of A
  - This A_k can be used, for example, for image compression

The Method

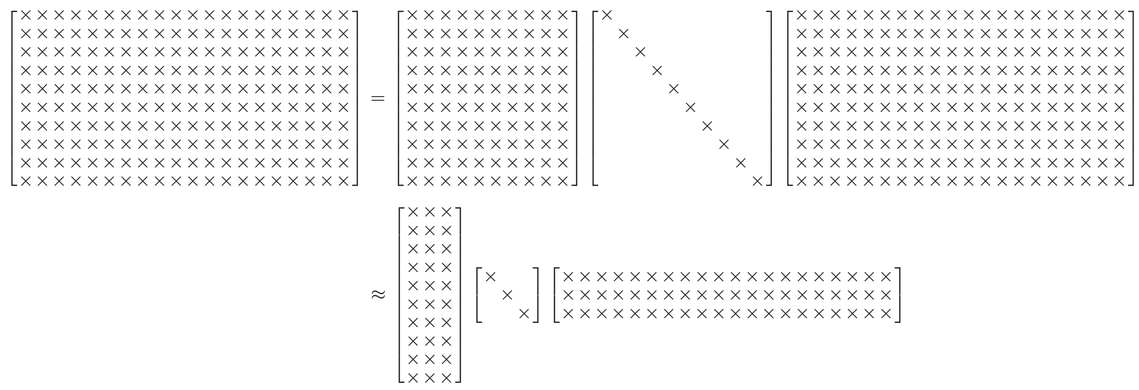
- The compression based on SVD works as follows.
  - Extract the RGB components as three matrices.
    For example, the **Red** matrix contains the **R**-values of every pixel.
  - For each matrix separately, we compute a truncated SVD.
  - Return the reduced factorizations as the RGB components of the compressed image

SVD Compression in a Nutshell

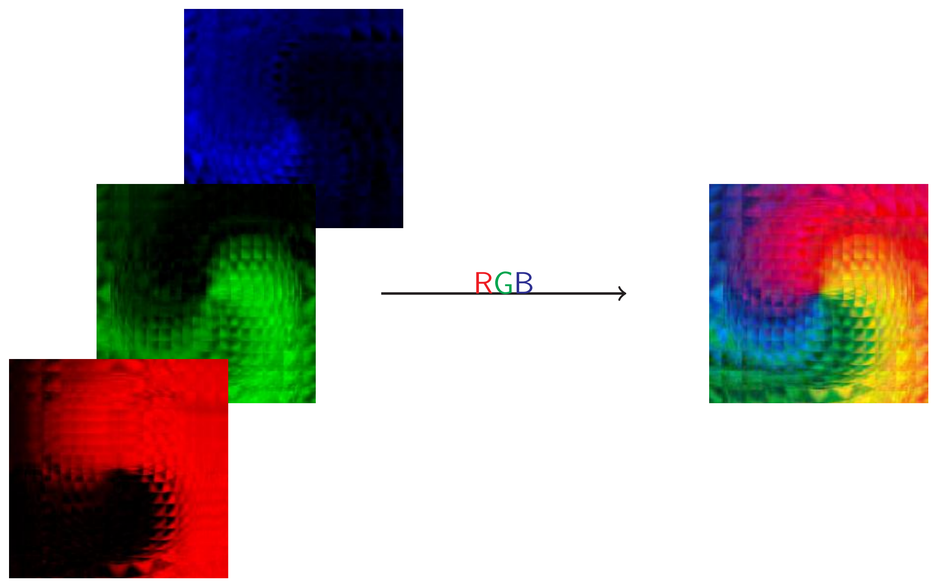
Extraction of the RGB components



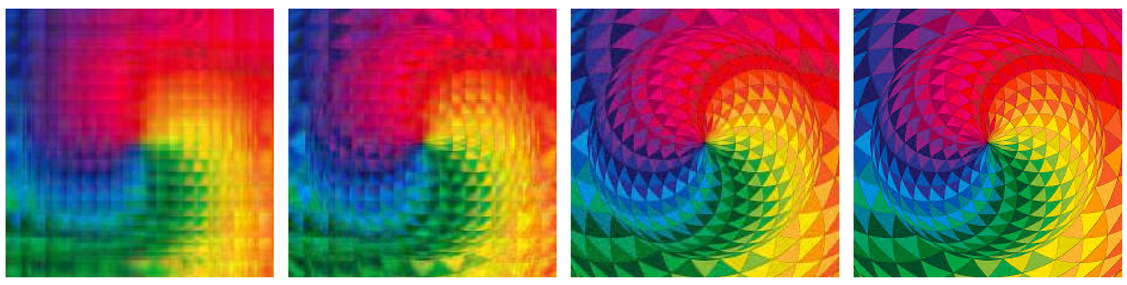
Truncated SVD



Reconstruction of the compressed image



Compression with different parameters



5 singular values	10 singular values	50 singular values	Original
2% of the data	4% of the data	18% of the data	100% of the data
16.8KB	33.6KB	168KB	900KB

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